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Paleomagnetic and paleointensity study of an Oligocene–Miocene lava sequence from the Hannuoba Basalts in northern China

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Abstract

A combined geochronologic, paleomagnetic and paleointensity investigation has been performed on a lava sequence from the Hannuoba Basalts in northern China. Potassium–argon (K–Ar) dating indicates that the lava sequence is of late Oligocene to middle Miocene age (~24–15 Ma). Rock magnetic measurements of thermomagnetic curves, hysteresis properties, saturation isothermal remanent magnetization (SIRM) and back-field demagnetization on representative samples from each lava flow in conjunction with reflected light microscopy observations show the primary carriers of remanence to be fine-grained pseudo-single domain titanomagnetite with various titanium contents. Stepwise thermal and alternating field demagnetization successfully isolated stable characteristic remanent magnetization from 40 out of the 43 lava flows, which record 24 paleomagnetically independent directions. Data (excluding directions regarded as abnormal/transitional) yield a mean direction of $D/I = 6.8^\circ/56.3^\circ$, with $\alpha_{95} = 5.2^\circ$ and $\kappa = 51.6$, corresponding to a paleomagnetic pole at 232.3°E, 83.7°N with $dp = 5.4^\circ$ and $dm = 7.5^\circ$, consistent with previously published paleomagnetic poles for the North China Block. The modified Thellier method with systematic partial thermoremanent magnetization (pTRM) checks and pTRM-tail checks was conducted on 72 selected samples. Reliable paleointensity estimates from four lava flows yield a mean virtual dipole moment of $(5.6 \pm 1.6) \times 10^{22}$ A m^2, suggesting moderately low intensity of the geomagnetic field. This is in good accord with reliable paleointensity datasets for the late Oligocene to middle Miocene (30–15 Ma), but lower than the average for middle Miocene to Pliocene age (15–2 Ma). It thus appears that at some point in the mid-Miocene there was a transition where the field intensity changes between these two states, although many more data are needed to confirm this.
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Keywords: Absolute paleointensity; Paleomagnetic direction; K–Ar dating; Oligocene–Miocene; Hannuoba Basalts; North China Block

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1. Introduction

Long-term variation in the direction and strength of the geomagnetic field is of considerable interest in geosciences because it is essential in understanding the processes of the Earth's dynamo over geological time (e.g. Prévot et al., 1990; Selkin and Tauxe, 2000; Kono and Roberts, 2002; Heller et al., 2002; Biggin and Thomas, 2003). In recent years, there has been a growing trend in documenting full-vector (both reliable paleodirectional and absolute paleointensity) variations of the geomagnetic field from volcanic sequences (e.g. Laj et al., 1999; Zhu et al., 2001, 2003; Goguitchaichvili et al., 2001, 2003; Riisager et al., 1999, 2002; Leonhardt and Soffel, 2002; Leonhardt et al., 2003). Recent studies have shown that the average field strength from the early Cenozoic to the Oligocene was about half of the present-day field (e.g. Riisager et al., 1999; Shi et al., 2002; Goguitchaichvili et al., 2003), whereas the average field strength is higher from Plio-Pleistocene to present. In the present global paleointensity database, there is only limited data of Miocene age that meet current high standard quality criteria. The exact timing and nature of the transition from low to high paleointensity still remains unknown.

The Hannuoba continental flood basalts are extensively distributed in northern China (Fig. 1). Previous potassium–argon (K–Ar) dating showed that the lava is of Oligocene–Miocene age, emplaced between 30 and 7 Ma (Liu et al., 1992). Zheng et al. (2002) reported K–Ar ages of 10–6, 33–22 and 31–17 Ma for the central and the eastern Hannuoba basalts (eastern Inner Mongolia and northern Hebei provinces). We have recently carried out a comprehensive paleomagnetic study in conjunction with new K–Ar dating on a lava sequence containing 43 lava flows at Ninglubao (NLB, 40°09.8′N, 112°41.6′E, see Fig. 1), one of the thickest outcrops in the southwestern Hannuoba basalts. Based on detailed rock magnetism and demagnetization analysis, paleointensity experiments were conducted on selected samples using the modified Thellier method (Coe, 1967) with sliding partial thermoremanent magnetization (pTRM) checks (Prévot et al., 1985) as well as pTRM tail checks (Riisager and Riisager, 2001) to monitor for alteration and multi-domain grain behavior.

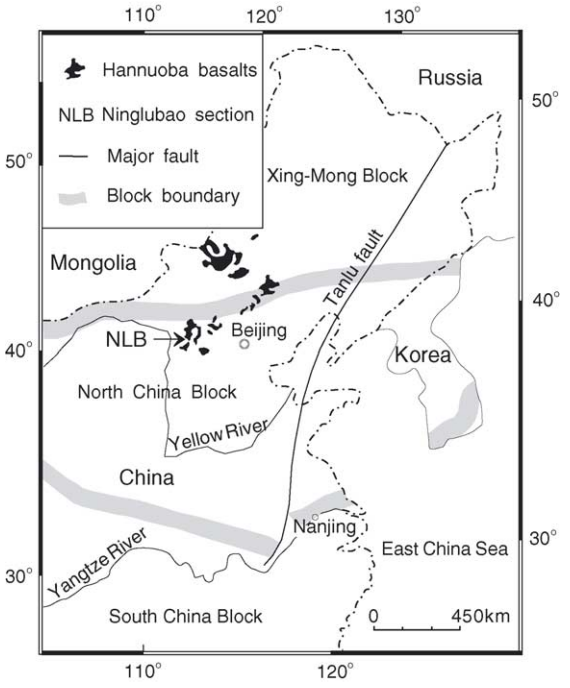


Fig. 1. Sketch map showing the distribution of the Hannuoba basalts and the sampling site location (Ninglubao, NLB).

2. Hannuoba basalts and sampling

The Hannuoba basalt has been regarded as the most intensive volcanic emplacement in northern China during the Cenozoic, which may relate to a regional rifting process (Liu et al., 1992). Lavas are presently distributed in Inner Mongolia, Heibei and Shanxi provinces in a basaltic highland with a height of hundreds of meters. Lithologically, the Hannuoba basalts consist of alkalic and tholeiitic basalts (Zhi et al., 1990; Chen et al., 2001). The tholeiitic and alkalic basalts are intercalated, usually with an alkalic basalt at the base of most profiles. Zhi et al. (1990) and others argued that the tholeiitic series originated from the lithospheric mantle and the alkalic series from the deeper asthenospheric mantle. The alkalic basalts host most deep-seated xenoliths while only very rare xenoliths have been found in the tholeiitic basalts. Liu et al. (2001) explained mafic and intermediate granulite xenoliths contained in Hannuoba basalts as products of fractional crystallization of a magma chamber in the lower crust. Moreover, previous paleomagnetic studies

have indicated that the whole North China Block has been tectonically stable since the Cretaceous (Zhao et al., 1990; Zheng et al., 1991; Zhu et al., 2001).

The NLB section is located in northern Shanxi province in the North China Block, with an altitude of 1639–1915 m. Lava flows are continuously exposed along the southern slope of a basalt highland. They consist of dark-grayish and black olivine-containing alkalic and tholeiitic basalts divided by violet or brown bubble horizons. Forty-three lava flows were recognized in field with flow thickness between 2 and 14 m. No sedimentary layers were found imbedded in this sequence; however, lavas #17, #25 and #35 are heavily weathered, indicating relatively longer intermittence between adjacent eruptions (Fig. 2). Horizontally layered flows indicate no significant tilting or tectonic movement since cooling.

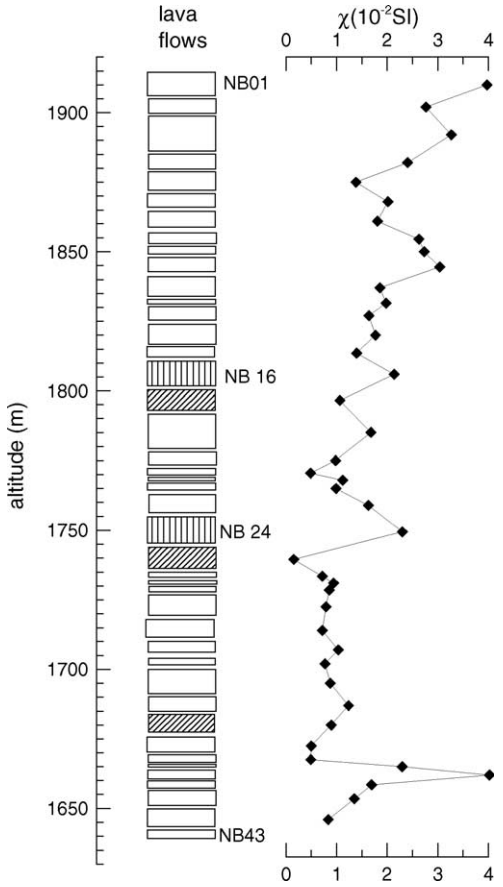


Fig. 2. Lava flow sequence and variation of low-field magnetic susceptibility. Vertical (inclined) shading boxes stand for columnar (strongly weathered) lava flows.

At least six paleomagnetic cores were drilled from each lava flow using a portable gasoline-powered drill. Cores were vertical distributed through the flow and oriented in situ by magnetic compass and/or sun compass. The difference between magnetic and sun compass readings was generally less than 6°. Two or three standard size paleomagnetic samples from each core were cut in the laboratory. In addition, fresh block samples were collected for radiometric dating.

3. Experiments and results

3.1. K–Ar dating

The K–Ar dating was conducted in the Geochronology Laboratory at the Institute of Geology, China Seismological Bureau. Fresh rock samples were crushed and sieved to a size of 0.20–0.28 mm. Olivine and plagioclase crystals that may contain excess argon were removed. Argon gas was analyzed using the isotope dilution technique with a 99.98% pure ^{38}Ar spike on a MM1200 mass spectrometer connected to a purification and extraction system. The spike was calibrated and corrected with standard argon minerals: B4M and ZBH. The decay and abundance constants used in the determinations are $\lambda\beta = 4.962\times10^{-10}\,a^{-1}$; $\lambda\varepsilon = 0.581\times10^{-10}\,a^{-1}$; $^{40}K/K = 1.167\times10^{-4}\,mol^{-1}$ (Steiger and Jager, 1977).

The K–Ar analytical data and resulting age determinations are presented in Table 1. Results indicate that the ages of lava flows #34, #14 and #03 are 23.9 ± 0.3, 16.1 ± 0.2 and 15.1 ± 0.6 Ma, respectively. It indicates that the entire sequence is dated from late Oligocene to middle Miocene age (the Oligocene–Miocene boundary age: 23.8 ± 1 Ma, Cande and Kent, 1995; 22.9 ± 0.1 Ma, Shackleton et al., 2000).

Table 1
K–Ar ages of basalts from the Nilubao (NLB) section

Lavas	K (%)	Ar (10^{-11} mol)	Age $\pm\delta$ (Ma)
NB03	1.634 ± 0.019	4.094 ± 0.112	15.1 ± 0.6
NB14	1.095 ± 0.010	3.073 ± 0.039	16.1 ± 0.2
NB34	0.650 ± 0.012	2.713 ± 0.038	23.9 ± 0.3

3.2. Rock-magnetic properties

Rock-magnetic studies, including room-temperature low-field magnetic susceptibility, saturation isothermal remanent magnetization (SIRM) acquisition, back-field demagnetization, hysteresis loops and temperature-dependence of magnetization (*J–T*) were carried out on at least two representative samples from each lava flow, in order to characterize the remanence carriers and to assess the thermal stability for the Thellier-type paleointensity experiments. Results are presented in Table 2.

Low-field magnetic susceptibilities were measured on one sample from each core using a Bartington MS-2 magnetic susceptibility meter. Lava flow mean susceptibilities vary from 15×10^{-4} to 403×10^{-4} SI, with a mean value of 160×10^{-4} SI. Susceptibility values increase from lava #24 upward (Fig. 2). Values are lower at the lower intervals except a high around lava #39.

SIRM, back-field demagnetization and hysteresis loops at room temperature were measured on a Magnetic Measurement Variable Field Translation Balance (MMVFTB). Coercive force (B_c) ranges from 4 to 36.9 mT, with an average of 17.3 mT; the coercivity of remanence (B_{cr}) changes between 10.1 and 65 mT, with an average of 32.3 mT. SIRM ranges between 5.6×10^{-3} and 427.4×10^{-3} A m^2 kg^{-1}. Ratios of hysteresis parameters, calculated after correction for the paramagnetic contributions, show that most samples fall into the pseudo-single-domain (PSD) range towards the single domain (SD) range on a Day-plot (Day et al., 1977) (Fig. 3). This indicates a mixture of multi-domain and a significant amount of single domain grains, with a large concentration of fine-grain particles.

Fig. 4 shows representative *J–T* curves in strong applied field (a steady field of 400 mT). Heating and cooling were in air atmosphere with rates of 8–10 °C min^{-1}. The Curie temperature was determined from heating curves by the inflection in the curve and as the intersection of the descending curve with the tangent to the subsequent paramagnetic tail. The thermomagnetic experiments allow a discrimination of four major types of thermomagnetic behavior.

Type A is characterized by a high Curie temperature (T_c) between 520 and 580 °C indicative of low-Ti titanomagnetite. The induced magnetization decays from room temperature to 450 °C, followed by a rapid decrease to the Curie point. The cooling curves are nearly reversible to the heating curves (Fig. 4a). Such thermomagnetic behavior suggests that negligible alteration of the magnetic mineralogy occurs during heating, which is essential in Thellier-type paleointensity experiments. Light microscope observations on polished thin sections confirmed that type-A samples show low-Ti titanomagnetite with ilmenite brim or lamella. Low-Ti titanomagnetite are usually formed by high temperature oxidation/exsolution of titanomagnetite into intergrowth of low-Ti magnetite and ilmenite during the original cooling of the lava.

Thermomagnetic curves of types B–D are irreversible (Fig. 4b–d). Specifically, types B and D are characterized by a low T_c (<300 °C) in the heating curve and distinct higher magnetization in the cooling curve (Fig. 4b and d). The increase in magnetization seen in the cooling curve suggests either an oxidation of high-Ti titanomagnetite/titanomaghemite into low-Ti magnetite and ilmenite or mineralogical phase transformations, e.g. new magnetic minerals formed from clay mineral alteration by heating. Type C is characterized by two T_c on the heating curves, 200–400 and 560–580 °C (Fig. 4c), probably indicating an inversion from titanomaghemite to magnetite during heating. Due to the tendency of thermal instability and the suspicion of low temperature alteration, types B–D were not used for the Thellier paleointensity experiments.

3.3. Demagnetization analysis

Stepwise thermal and alternating field (AF) demagnetization analyses were conducted at the paleomagnetism laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences (IGG-CAS). Remanent magnetizations were measured with a 2G cryogenic magnetometer and an AGICO JR-5a spinner magnetometer in a shielded room (<300 nT). Thermal demagnetizations of the natural remanent magnetization (NRM) were made on a MMTD60 thermal demagnetizer with residual field less than 10 nT. AF demagnetization and remanent measurements were conducted on a 2G cryogenic magnetometer with built-in AF demagnetizer. Thermal demagnetizations were conducted up to a peak temperature of 585 °C with temperature steps from 25 to 50 °C, while AF demagnetizations were up to a maximum field of 80 mT at

Table 2
Rock-magnetic properties of basalts from the NLB section[a]

Lava	H (m)	χ (10^{-4} SI)	B_c (mT)	B_{cr} (mT)	M_s (10^{-3} A m^2 kg^{-1})	SIRM/χ (10^{-3} A m^2 kg^{-1})	SIRM/κ	T_c	T_c type
NB 01	1915	397.7	22.0	46.6	359.0	45.0	1.13	576	A
NB 02	1905	276.8	14.5	29.3	1112.6	31.8	1.15	350/510	C
NB 03	1899	327.0	7.8	12.3	1330.9	75.0	2.29	230	B
NB 04	1885	240.6	8.7	10.3	1060.2	36.0	1.50	218	B
NB 05	1879	138.2	12.8	30.3	778.1	143.4	10.38	256	D
NB 06	1871	201.5	17.5	36.0	841.0	111.8	5.55	573	A
NB 07	1865	180.7	11.9	21.1	614.6	57.5	3.18	563	A
NB 08	1857	262.7	14.9	28.8	751.9	71.9	2.74	573	A
NB 09	1852	273.0	33.2	54.7	1169.7	427.4	15.66	566	A
NB 10	1848	304.0	25.1	61.3	1391.1	277.6	9.13	568	A
NB 11	1841	185.5	23.3	37.2	1672.2	254.8	13.74	192	B
NB 12	1833	197.9	7.4	12.8	1274.5	190.0	9.60	310/480	D
NB 13	1830	163.7	20.6	38.8	1309.3	157.7	9.63	558	A
NB 14	1824	176.7	16.6	35.6	1440.0	140.0	7.92	570	A
NB 15	1816	139.5	18.6	41.0	1341.9	215.6	15.45	300/560	C
NB 16	1811	213.5	17.8	34.3	1309.6	148.4	6.95	586	A
NB 17	1801	106.4	21.6	37.1	600.0	56.0	5.26	160	B
NB 18	1792	168.0	8.2	10.7	537.1	20.0	1.19	312/575	C
NB19	1778	–	–	–	–	–	–	–	–
NB 20	1772	48.4	6.8	14.7	242.6	9.0	1.86	161	B
NB 21	1769	112.1	9.1	19.0	1165.3	78.7	7.02	565	A
NB 22	1767	98.8	15.2	24.4	502.0	48.4	4.90	563	A
NB 23	1763	162.6	4.0	13.7	283.4	5.6	0.34	–	–
NB24	1755	15.3	15.9	17.2	362.6	–	–	570	A
NB 25	1744	–	–	–	–	–	–	–	–
NB 26	1735	71.5	12.0	15.1	362.6	41.7	5.83	578	A
NB 27	1732	94.0	8.9	13.3	277.0	11.0	1.17	430/562	C
NB 28	1730	85.3	22.8	54.1	460.1	66.8	7.83	150/575	C
NB 29	1727	79.0	36.9	65.0	365.9	75.7	9.58	586	A
NB 30	1718	71.5	25.6	47.3	288.0	43.4	6.06	590	A
NB 31	1710	103.2	35.8	57.9	426.7	100.6	9.75	558	A
NB 32	1704	77.2	26.4	54.2	499.1	89.8	11.63	582	A
NB 33	1700	87.5	31.3	55.4	358.3	100.6	11.50	580	A
NB 34	1690	123.5	7.6	10.1	416.1	8.6	0.70	220	B
NB 35	1684	89.3	4.8	20.3	1454.7	61.6	6.89	210	B
NB 36	1676	49.6	15.9	12.7	190.5	17.3	3.49	218	B
NB 37	1669	48.9	15.8	17.1	130.2	10.2	2.09	225	B
NB 38	1666	229.5	7.6	10.1	416.1	8.6	0.38	270	B
NB 39	1664	402.5	4.5	18.7	1364.8	64.2	1.59	245	B
NB 40	1660	169.2	7.8	18.1	590.1	36.0	2.13	557	A
NB 41	1657	135.0	21.8	47.2	187.0	37.3	2.76	170	D
NB 42	1650	83.1	21.2	46.1	206.1	34.4	4.14	564	A
NB 43	1642	108.2	21.8	33.8	390.2	92.4	8.54	450/586	C

[a] H, altitude of the upper boundary of lava flow; χ, low-field magnetic susceptibility; B_c and B_{cr}, the coercive force and coercivity of remanence; M_s, saturation magnetization; SIRM/χ, saturation isothermal remanent magnetization; T_c, Curie temperature.

incremental steps of 5–10 mT. Thermal and AF demagnetization were found to be equally effective in determining the characteristic remanent magnetization (ChRM).

Demagnetization analyses yielded univectorial ChRMs (Fig. 5). Generally, a small magnetization of viscous origin was present and easily removed by 150–200 °C in thermal treatment and by 10–15 mT in AF demagnetization. A stable ChRM was obtained from 260 out of 279 measured samples (93%). The ChRM directions were calculated using principal component analysis of Kirschvink (1980). Lava-mean and

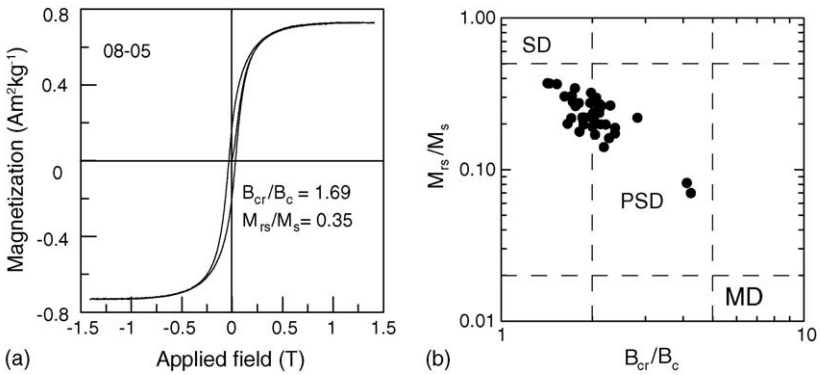


Fig. 3. Representative room-temperature hysteresis loop (a) and Day-plot of measured samples (b). M_{rs}/M_s, ratio of saturated remanence to saturation magnetization; B_{cr}/B_c, ratio of coercivity of remanence to coercive force. SD, PSD and MD refer to single domain, pseudo-single domain and multi-domain, respectively.

group mean directions were calculated using Fisher statistics (Fisher, 1953). Forty out of 43 lava flows yielded lava-mean directions. Results are summarized in Table 3 and presented in Fig. 6. It was not possible to obtain flow mean directions for three lava flows, lavas #02, #13 and #25. For lavas #02 and #13 it was not possible to isolate the ChRM for some individual samples, and from those where it was possible there was a very large scatter in directions within the flow indicating that the flows were possibly affected by lightening. No samples from the flow #25 were collected because it is too strongly weathered to sample in the field.

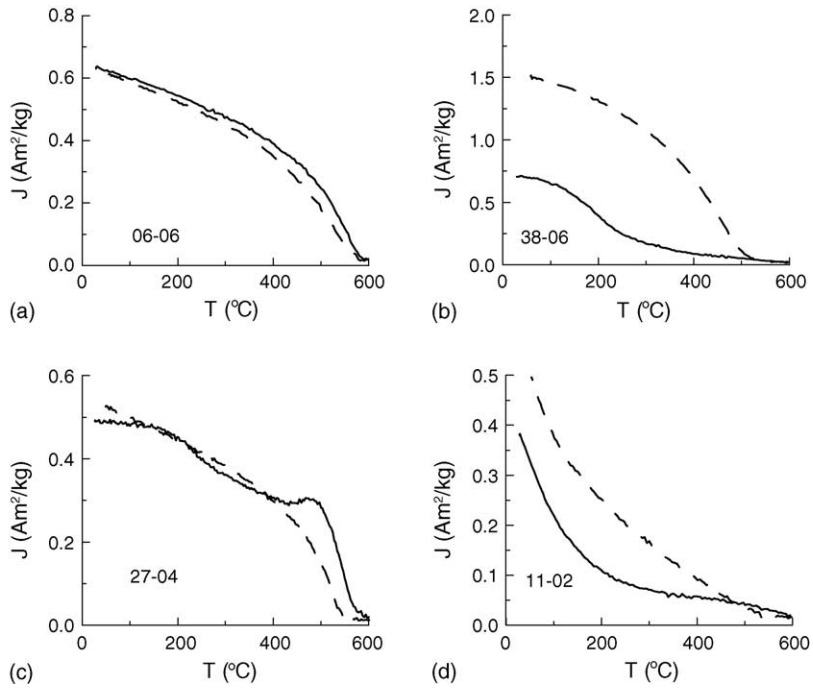


Fig. 4. Representative strong applied field (400 mT) thermomagnetic curves of magnetization versus temperature (*J–T*). Samples were heated in air at a rate of 20 °C min^{-1}. Solid (dashed) line corresponds to heating (cooling) curve. Type A samples were selected for Thellier paleointensity experiments.

Table 3
Individual lava mean directional results (left) and grouped mean results (right) from the studied section

Lavas	H (m)	n/N	D (°)	I (°)	κ	α_{95} (°)	Plat (°)	Plong (°)	DG	Number	D (°)	I (°)	κ	α_{95} (°)	Plat (°)	Plong (°)
NB01	1915	6/7	174.5	−55.8	70	8.1	−84.2	162.9								
NB02	1905	–	–	–	–	–	–	–	DG01	14	179.4	−56.2	103	3.9	−86.6	120.7
NB03	1899	4/4	183.8	−58.7	247	5.9	−87	35.5								
NB04	1885	4/5	182.3	−54.1	128	8.1	−84.2	93.7								
NB05	1879	9/9	189.0	−60.6	123	4.7	−83.1	8	DG02	17	189.3	−59.6	159	2.8	−82.9	17.5
NB06	1871	8/8	189.7	−58.6	216	3.8	−82.5	26								
NB07	1865	8/8	191.8	−52.1	79	6.3	−78	57.2	DG03	8	191.8	−52.1	79	6.3	−78	57.2
NB08	1857	9/9	188.9	−60.3	140	4.4	−83.2	10.8								
NB09	1852	11/11	181.4	−59.0	89.8	4.8	−88.9	42.6	DG04	35	182.2	−59.5	102	2.4	−88.3	16.5
NB10	1848	5/6	182.9	−60.8	91	8.1	−87.3	344.7								
NB11	1841	10/10	176.8	−58.5	102	4.8	−87.4	182.6								
NB12	1833	4/4	199.0	−5.0	88	9.9	−48.6	83.2	DG05	4	199.0	−5.0	88	9.9	−48.6	83.2
NB13	1830	–	–	–	–	–	–	–								
NB14	1824	3/5	230.0	−40.3	272	7.5	−44.8	29.4	DG06	3	230.0	−40.3	272	7.5	−44.8	29.4
NB15	1816	4/5	330.1	34.1	155	7.4	56.5	351.6	DG07	4	330.1	34.1	155	7.4	56.5	351.6
NB16	1811	12/13	21.4	15.8	154	3.5	52.6	256.2	DG08	12	21.4	15.8	154	3.5	52.6	256.2
NB17	1801	5/5	187.3	−48.7	62	9.8	−77.9	80.9	DG09	5	187.3	−48.7	62	9.8	−77.9	80.9
NB18	1792	4/5	174.4	−24.9	210	6.4	−62.5	124.6	DG10	4	174.4	−24.9	210	6.4	−62.5	124.6
NB19	1778	4/4	30.0	55.1	103	9.1	66	203.9	DG11	4	30.0	55.1	103	9.1	66	203.9
NB20	1772	4/5	171.0	−64.3	104	9.1	−81.2	247.8								
NB21	1769	10/10	176.2	−59.1	39	7.8	−87.1	198.3	DG12	28	170.6	−60.9	62	3.5	−82.7	219.8
NB22	1767	5/5	162.8	−61.6	75	8.9	−76.9	219.7								
NB23	1763	9/9	168.3	−60.9	90	5.5	−81	217.9								
NB24	1755	7/8	163.7	−73.3	82	6.7	−68.5	269.5	DG13	7	163.7	−73.3	82	6.7	−68.5	269.5
NB25	1744	–	–	–	–	–	–	–								
NB26	1735	3/3	34.0	60.9	484	5.6	64.5	187.5								
NB27	1732	5/5	36.1	62.3	228	5.1	63.1	183.4	DG14	33	33.7	61.0	195	1.8	64.7	187.3
NB28	1730	7/7	32.5	62.7	238	3.9	65.7	182.6								
NB29	1727	7/7	36.9	61.6	132	5.3	62.4	185.1								
NB30	1718	11/11	31.6	59.0	183	3.4	65.9	193.3								
NB31	1710	5/5	29.9	56.3	135	6.6	66.5	201.1	DG15	13	28.6	58.3	151	3.4	68	196.4
NB32	1704	8/9	27.7	59.5	158	4.4	68.9	193.2								
NB33	1700	9/9	33.6	63.4	170	4.0	65	180.5	DG16	9	33.6	63.4	170	4.0	65	180.5
NB34	1690	6/7	44.2	55.6	242	4.3	55.4	194.9	DG17	6	44.2	55.6	242	4.3	55.4	194.9
NB35	1684	3/5	171.6	−39.9	170	9.5	−71.1	137.3	DG18	3	171.6	−39.9	170	9.5	−71.1	137.3
NB36	1676	3/5	195.7	−37.9	433	5.9	−66.9	72.8	DG19	7	181.8	−45.7	19	14.4	−76.9	105.6
NB37	1669	4/5	179.9	−47.6	160	7.3	−78.5	113.1								
NB38	1666	5/5	15.7	67.0	115	7.2	75.4	156.8	DG20	11	13.2	66.0	114	4.3	77.5	157.4
NB39	1664	6/6	11.3	65.1	100	6.7	79.3	158.5								
NB40	1660	8/8	8.5	49.8	82	6.2	78.2	254.2	DG21	8	8.5	49.8	82	6.2	78.2	254.2
NB41	1657	3/4	−13.7	64.5	605	5.0	78.3	59.1	DG22	3	−13.7	64.5	605	5.0	78.3	59.1
NB42	1650	14/15	9.3	46.5	184	2.9	75.4	258	DG23	14	9.3	46.5	184	2.9	75.4	258
NB43	1642	8/8	24.8	54.2	79	6.3	69.7	210.3	DG24	8	24.8	54.2	79	6.3	69.7	210.3

n/N, the number of samples used for calculation of the direction out of number of samples measured; D and I, mean declination and inclination of ChRMs; κ and α_{95}, the precision parameter and the confidence cone; Plat and Plong, the latitude and longitude of the virtual geomagnetic pole (VGP); DG, directional group.

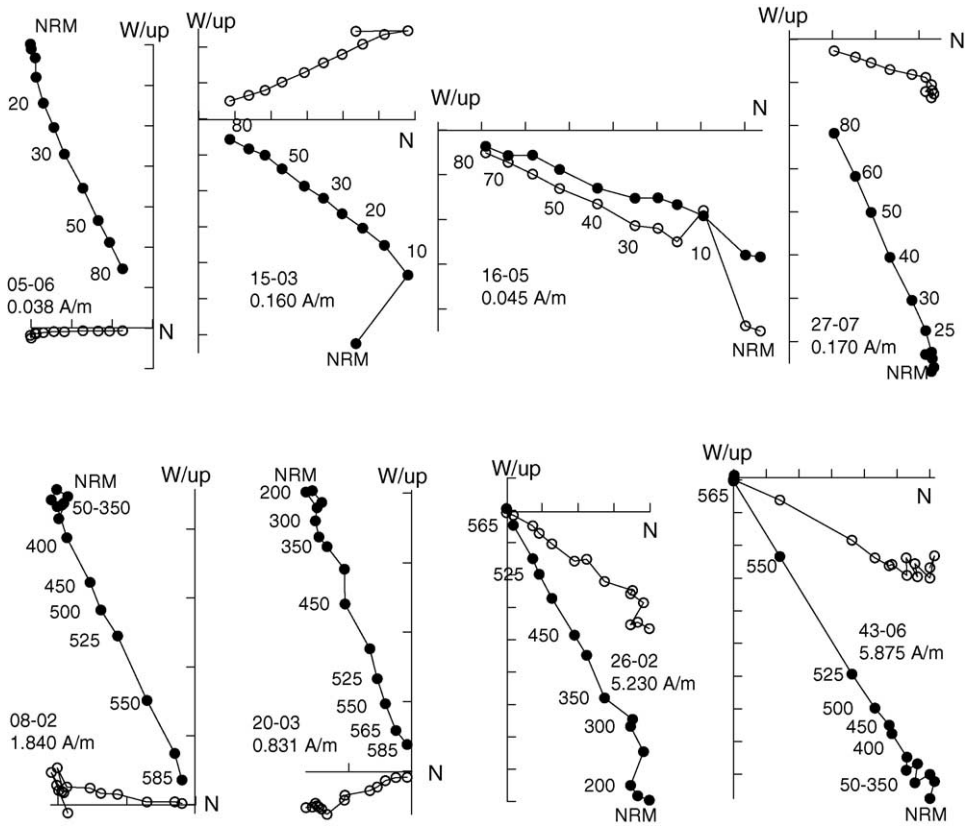


Fig. 5. Orthogonal vector plots of stepwise thermal (bottom row) and alternating field (top row) demagnetization in stratigraphic coordinates. The numbers refer to temperature in °C and peak field in mT, respectively. Solid (open) circles are projections into vertical (horizontal) planes.

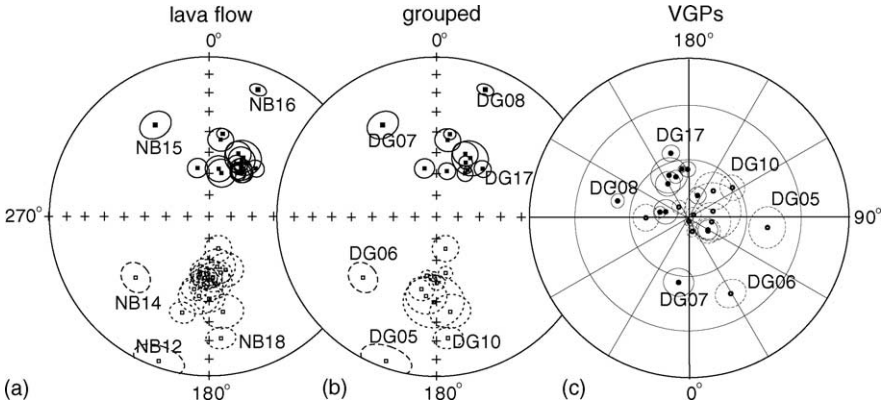


Fig. 6. Equal-area projections of mean directions with 95% confidence limits for (a) mean directions of individual lava flow; (b) directional groups (DGs). Solid (open) squares represent downward (upward) inclinations. (c) Corresponding virtual geomagnetic poles (VGPs) of the DGs in (b). Solid (open) circles stand for normal and reversed polarities.

3.4. Directional groups (DGs) analyses

As the samples were collected from successive lava flows in a vertical sequence, a bias in the statistical analysis may be introduced by giving too much weight to a single direction recorded by successive lava flows emplaced in a short time interval. In order to quantify whether the sampled lava flows recorded independent states of the geomagnetic field directions from successive flows were compared. The probability of independence between successive lava flows using the *F*-distribution test (e.g. Butler, 1992) was calculated. For values less than 95%, lava flows were assumed to have recorded the same geomagnetic field and subsequently combined in to directional groups (DGs). Nevertheless, it is also important to keep in mind that the geomagnetic field itself may stay stable for a significant period of time (compared to the rate of secular variation) so that successive lava flows that were not emplaced rapidly may record similar directions (Love, 2000). In such a case, grouping the data will overestimate the variability of the paleofield (Plenier et al., 2003). As shown in Fig. 6b and Table 3, a total of 24 DGs were obtained by the independence test.

From the base of the section upward a N–R–N–R–(N?)–R (N, normal and R, reversed) polarity structure is seen. Taking into account the new K–Ar ages, this corresponds to polarity chrons C7–C5 of late Oligocene to middle Miocene time. Some of the observed declinations and inclinations however, differ from those expected at this sampling site. For example, DGs 05 to 08 yield very shallow to moderate inclinations and very deflected declinations. Note that each of these DGs corresponds to a single cooling unit. Previous paleomagnetic investigations from volcanic and sedimentary rocks indicate that there has been no significant rotation or displacement in the North China Block since the Cretaceous or late Jurassic (e.g. Zhao and Coe, 1987; Zhao et al., 1990; Zhu et al., 2001). Our field observations at the sampling section and adjacent areas also show no local tectonic deformation leading us to be confident that the directional data is exempt from any tectonic influence. We thus conclude that these directional anomalies are related to the geomagnetic field (e.g. excursions) during lava emplacement. However, it is premature at this stage to discuss further the possible transitional field behavior recorded in this profile, as more detailed chronologic and paleomagnetic data are needed.

To avoid the influence of the transitional field, we tentatively designate directions with virtual geomagnetic pole (VGP) latitude less than 65° as transitional and not representative of the stable geomagnetic field (Fig. 6c). For the remaining 16 DGs, the seven normal ones yield a mean direction, $D/I = 15.2°/57.1°$ ($\alpha_{95} = 8.0°$) and the nine reversed ones yield a mean direction of $D/I = 180.6°/-55.4°$ ($\alpha_{95} = 7.0°$). The reversal test (McFadden and Lowes, 1981) for the 16 DGs is positive and application of McFadden and McElhinny (1990) classification of the reversal test shows a class C test ($\gamma_0 = 8.3°$) with a critical angle of $\gamma_c = 10.1°$ at the 95% confidence level. Consequently, the overall mean direction is $D/I = 6.8°/56.3°$, with $\alpha_{95} = 5.2°$ and $\kappa = 51.6$, corresponding to a paleomagnetic pole of 232.3°E, 83.7°N, with dp = 5.4° and dm = 7.5°. This is consistent with other Miocene paleomagnetic poles obtained from the North China Block (e.g. Zhao and Coe, 1987; Zheng et al., 1991; Zhao et al., 1994; Huang et al., 2004) and the Eurasia pole from the master apparent polar wander curve (Besse and Courtillot, 2002).

3.5. Paleointensity experiments

The Thellier-type paleointensity experiments were carried at the IGG-CAS. To maximize successful intensity estimations, only Type A samples were chosen for paleointensity experiments, as samples showing a single high Curie temperature, indicative of high temperature oxidation during original cooling of the lava, are usually considered to be stable under laboratory heating. Additionally, only samples from lava flows that yielded a flow mean direction by demagnetization analyses were selected. Thus, 72 samples (18 lava flows, 4 samples from each) were preselected for paleointensity experiment.

The Coe version of the Thellier method (Coe, 1967) was used on the selected samples with sliding pTRM checks (Prévot et al., 1985) as well as pTRM-tail checks (Riisager and Riisager, 2001) to check for alteration and/or multi-domain behavior. At each temperature step, the samples were heated twice: in zero-field for the first heating and in a laboratory applied field for the second heating. In this study, a steady laboratory field of 29 μT, parallel to the *z*-axis of samples, was applied during both heating and cooling cycles for each

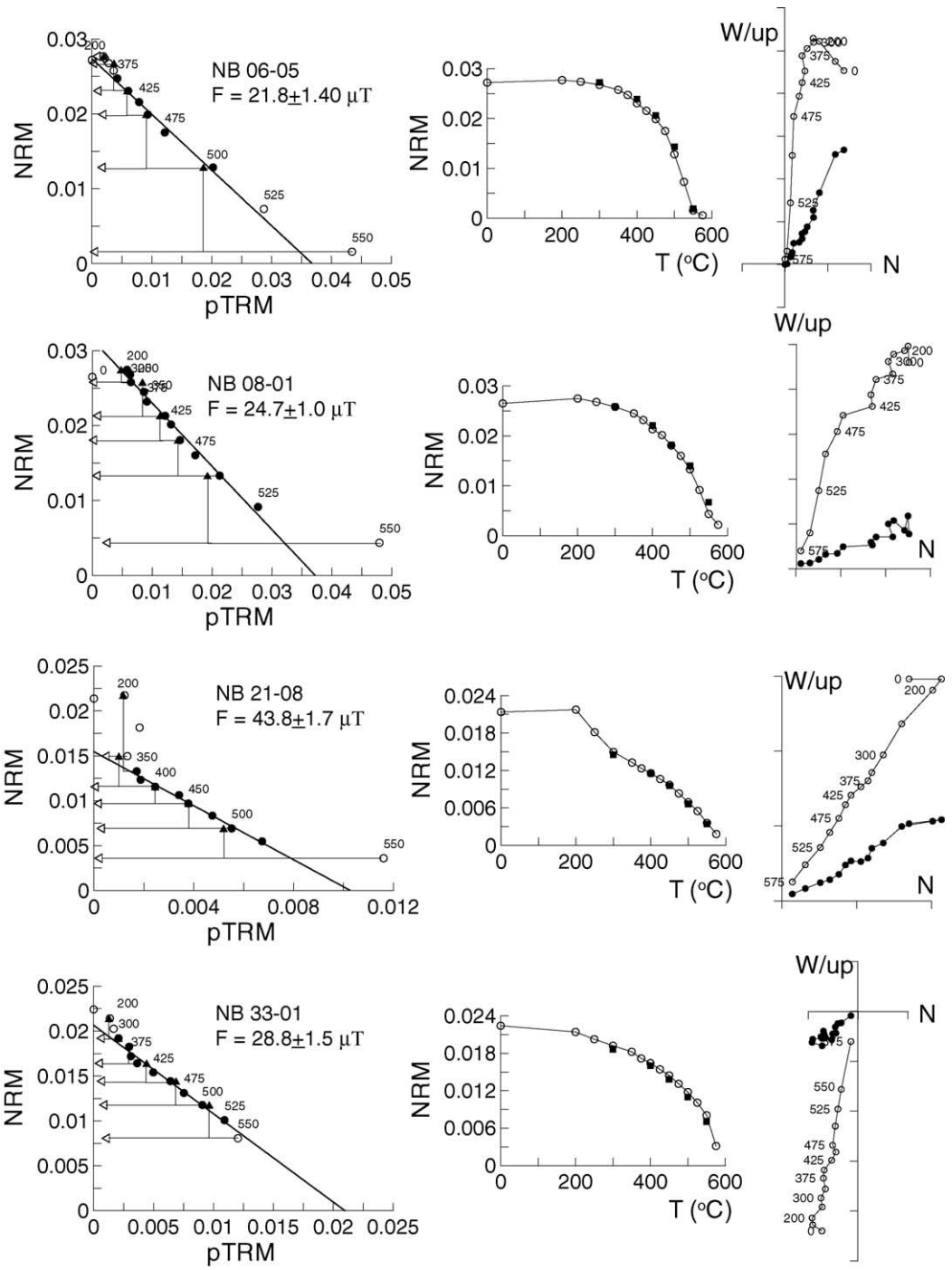


Fig. 7. Examples of accepted NRM–TRM plots and corresponding NRM demagnetization behavior (intensity and direction in an OVP plot). NRM and TRM in units of $A\,m^2\,kg^{-1}$. Solid (open) circles on the TRM-NRM plots are accepted (rejected) points and solid triangles are the partial thermal remanent magnetization (pTRM) checks. The pTRM-tail checks are plotted on the TRM axis on NRM–TRM plots with open triangles and solid squares on the NRM demagnetization intensity plots. Solid (open) circles on orthogonal projections are vertical (horizontal) planes in sample coordinates.

Table 4
Paleointensity results[a]

Site	Sample	T_{min}–T_{max} (°C)	n	f	g	q	$F \pm$ S.D. (μT)	VDM (10^{22} A m^2)	$VDM_e \pm$ S.D. (10^{22} A m^2)
NB06 (DG02)	06–05	375–500	6	0.44	0.72	4.8	21.8 ± 1.4	3.75	**4.3 ± 0.4**
	06–07	200–500	10	0.60	0.79	6.0	25.3 ± 2.0	4.35	
	06–08	250–475	8	0.40	0.82	5.5	26.1 ± 1.6	4.49	
	06–09	250–475	8	0.43	0.82	7.0	27.4 ± 1.4	4.71	
NB08 (DG04)	08–01	200–525	11	0.59	0.85	12.7	24.7 ± 1.0	4.25	**4.6 ± 0.3**
	08–02	200–525	11	0.47	0.86	10.9	28.6 ± 1.1	4.93	
	08–09	450–525	4	0.30	0.62	4.9	27.6 ± 1.1	4.75	
NB21 (DG12)	21–07	400–525	6	0.36	0.78	3.3	49.4 ± 4.2	8.35	**7.9**
	21–08	350–525	8	0.50	0.84	10.8	43.8 ± 1.7	7.41	
NB33 (DG16)	33–01	300–525	9	0.43	0.86	7.1	28.8 ± 1.5	4.71	3.6
	33–07	375–500	6	0.34	0.71	2.9	15.4 ± 1.3	2.52	
NB42 (DG23)	42–02	375–550	8	0.35	0.82	6.2	28.7 ± 1.3	5.77	**5.4**
	42–07	375–525	7	0.48	0.82	6.7	24.6 ± 1.4	4.95	
Average									**5.6 ± 1.6**

[a] T_{min}–T_{max} specify the temperature range of NRM–TRM straight-line segment used for paleointensity determination; n, number of points used; f, g and q are the fraction of NRM used, the gap factor, and quality factor, respectively (Coe et al., 1978); $F \pm$ S.D., paleointensity estimate with standard deviation; VDM and VDM_e, individual and average virtual dipole moments. Only those VDM_e in bold (so excluding NB33) have been used to determine the overall average (see text).

in-field treatment. At every other temperature step (after 300 °C) samples were heated a third time to the same temperature in zero-field in order to determine the pTRM tail. Then the fourth heating was to a lower temperature in the applied field in order for a pTRM check to be obtained.

Thermal treatment steps were at 100, 200, 250, 300, 350, 375, 400, 425, 450, 475, 500, 525, 550 and 575 °C. Samples were treated in a paleointensity furnace using the following procedure. During heating a rate of 3.2 °C min^{-1} was applied until the temperature in the furnace reached 60 °C below the required peak temperature at which point a slower rate of 1 °C min^{-1} was used until the peak temperature was attained. Samples were then held at temperature for 30 min in order for thermodynamic equilibrium to be reached. The samples were then left to cool naturally overnight. In order to minimize chemical alteration during heating prior to each run air was first evacuated from the furnace with a primary vacuum pump, then argon was alternatively flushed in and pumped out of the chamber at least three times. Argon gas was circulated gently through the furnace during the whole experimental process. In addition, a few milligrams of activated charcoal, a buffer to any oxidation/reduction of magnetite, was introduced into the furnace prior to each run (Laj and Kissel, 1999).

The NRM remaining after each thermal run has been plotted against the corresponding thermal remanent magnetization (TRM) acquired in the applied laboratory field at the same peak temperature. In an ideal diagram, the NRM–TRM points fall onto a straight line, the slope of which gives the ratio of the laboratory field to the paleofield. The quality of each paleointensity determination has been evaluated using the statistical parameters developed by Coe et al. (1978). We accepted only determinations that satisfied all of the following requirements: (1) at least four NRM–TRM points used for paleointensity determination, which corresponds to an NRM fraction $f > 0.3$, and the NRM directions obtained from the paleointensity experiment heads towards the origin with MAD < 6°; (2) the quality factor q is larger than 2 and the standard error of the slope/slope (β factor) is <0.1; (3) positive pTRM checks (within 10% difference normalized to the total TRM), and (4) positive pTRM-tail checks (less than 6% difference normalized to the NRM).

Thirteen out of 72 samples from 5 out of 18 independent lava flows passed the above quality criteria. Examples of accepted NRM–TRM diagrams are shown in Fig. 7 and paleointensity results are summarized in Table 4. They yielded reliable individual paleointensity determinations with virtual dipole moments (VDMs) ranging from 2.52×10^{22} to 8.35×10^{22} A m^2. De-

spite preselection being applied, only ∼18% samples yielded acceptable paleointensity estimates. Alteration during the experiment was the main cause of failure so that criteria (2) and (3) were often not fulfilled. Despite the Curie curves indicating stability on heating and an argon atmosphere in the oven during heating the samples were not stable to the prolonged heating (with two or four heatings at each temperature) during the palaeointensity experiment.

The two estimates for lava NB33 are not consistent and so are not considered to accurately represent the field intensity. Two flow estimates (NB06 and NB08, DG02 and DG04) correspond to ages less than 16.1 Ma, whilst the estimate for NB42 (DG23) has an age greater than 24 Ma. However, as there is only limited data the eleven estimates from the four flows are grouped to give an average VDM of $(5.6 \pm 1.6) \times 10^{22}$ A m^2 for the time interval 24–15 Ma.

4. Discussion and conclusions

The new K–Ar ages indicate that volcanic eruptions took place at northern Shanxi province from

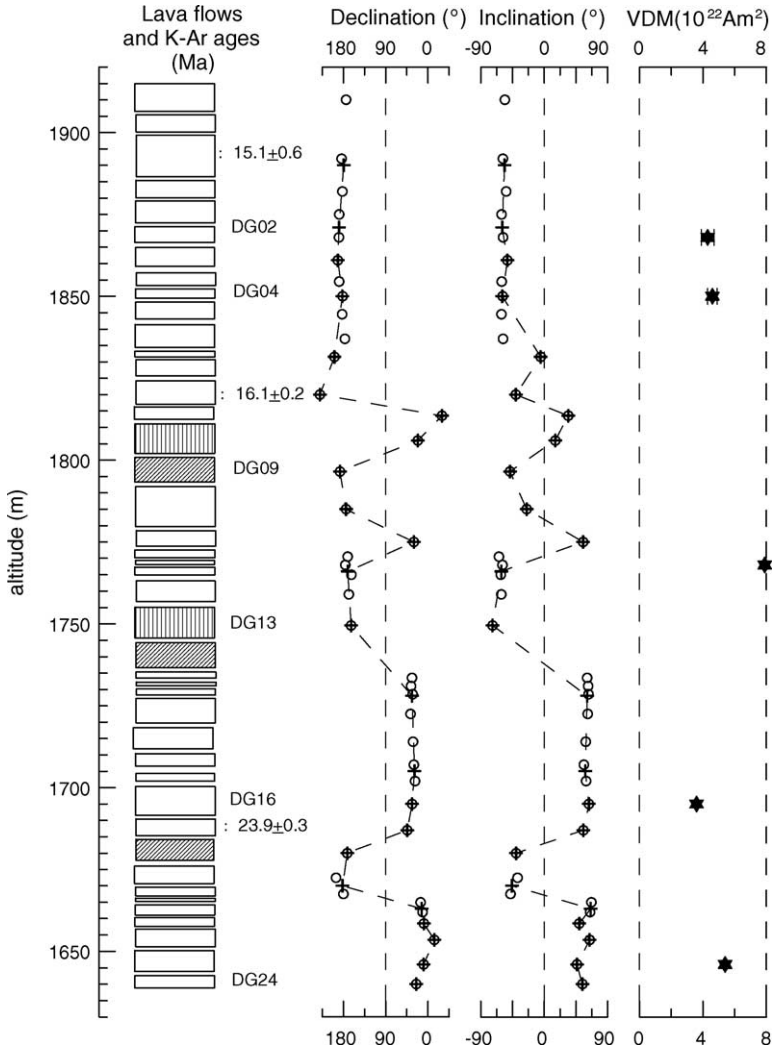


Fig. 8. Declination, inclination, and virtual dipole moment (VDM) of the NLB section as a function of altitude. For the directional results, both individual lava flow (open circles) and the grouped directions (DGs, crosses) are plotted.

late Oligocene to middle Miocene from ~24 to 15 Ma (the difference between ages of lavas #34 and #03). Strongly weathered lavas #17, #25 and #35 indicate that there were periods of time where no eruptions took place and that the eruptions may have been intermittent. Zheng et al. (2002) proposed that a volcanic hiatus between 25 and 10 Ma occurred in eastern Inner Mongolia. Contrary to this, our K–Ar ages indicate that magma was active during the middle Miocene in the northern Shanxi province (western Hannuoba). This may be possible to track the spatial temporal migration characteristics of magma in the North China Block during this time. However, uncertainties in the K–Ar dating must also be considered. Uncertainties could be introduced by undetected alteration of the rocks. A low K content (possibly indicating K loss) in lava #34 might suggest an overestimation of age. $^{40}Ar/^{39}Ar$ dating is planned for more detailed age determinations.

Directional, paleointensity and age results are shown in Fig. 8 for the whole section. Assuming the new K–Ar dates are accurate, we conclude that the magnetostratigraphy may correlate to polarity chrons C7–C5 (Cande and Kent, 1995). The paleomagnetic pole of 232.3°E, 83.7°N (with dp = 5.4° and dm = 7.5°) of the NLB section is consistent with other Miocene paleomagnetic poles obtained from the North China Block, and the Eurasia pole from the master apparent polar wander curve (Besse and Courtillot, 2002). This further confirms that the magnetization is primary and there has been no significant tectonic movement in this region since the middle Miocene eruptions. Furthermore, the high-quality paleointensity data obtained in this study can be compared to the present global paleointensity data set for the Neogene period. We consider data in the existing global palaeointensity database (PINT2003, ftp://dstu.univ-montp2.fr/pub/paleointdb/) that satisfy the following reliability criteria: (1) the determinations were obtained through some form of the stepwise Thellier method with positive pTRM checks; (2) at least two successful determinations per cooling unit and a minimum of nine estimates from at least three cooling units were used to determine the mean VDM; (3) the standard error of the mean for a given cooling unit must be less than 20%; (4) no data were considered from a transitional field; (5) datasets have radiometric dating constraints. In addition to the PINT2003 datasets, we also include recent datasets for the Pliocene from Fernando Noronha, Brazil (Leonhardt et al., 2003). In total, 12 independent records meet the selection criteria. The selected mean VDMs, along with the result from this study are shown in Fig. 9 as a function of age.

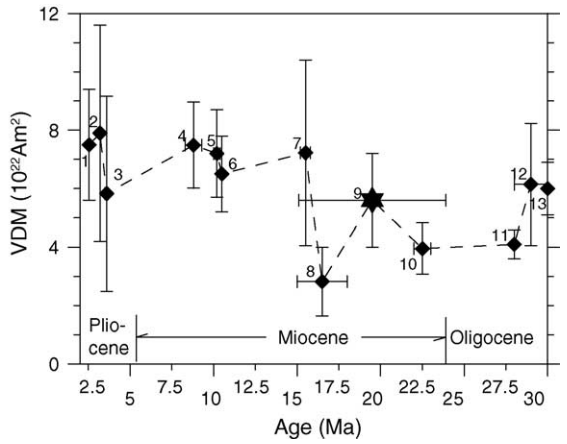


Fig. 9. Changes of mean virtual dipole moments from the late Oligocene to Pliocene from this study (star) and selected data (diamonds) (see text). Numbers refer to the particular study: (1) Leonhardt et al. (2003); (2) Laj et al. (2000); (3 and 4) Goguitchaichvili et al. (2000a, 2002); (5 and 6) Riisager et al. (2000); (7) Prévot et al. (1985); (8) Goguitchaichvili et al. (2000b); (9) this study; (10) Shi et al. (2002); (11) Goguitchaichvili et al. (2001); (12) Plenier et al. (2003); (13) Riisager et al. (1999).

It can be seen that the mean VDM from the present study, $(5.6 \pm 1.6) \times 10^{22}$ A m^2 for the time interval 15–24 Ma, is comparable to the average values for the Oligocene to early Miocene obtained from Ethiopian flood basalts (Riisager et al., 1999), the southern Indian ocean (Plenier et al., 2003), Mexican volcanics (Goguitchaichvili et al., 2000a, 2001) and Pingzhuang basalts, China (Shi et al., 2002). These studies yield an average VDM value of $(4.6 \pm 1.4) \times 10^{22}$ A m^2 for the late Oligocene to early Miocene, which with our new data included becomes $(4.8 \pm 1.3) \times 10^{22}$ Am^2. This mean is lower than the mean VDM for the late Miocene to Pliocene (15–2 Ma). The average VDM for the more recent interval is $(7.1 \pm 0.7) \times 10^{22}$ Am^2, which is ~48% higher than that for the late Oligocene to early Miocene interval. This suggests that there is a transition between the low and the high field modes during the mid-Miocene, although further data are needed to confirm this.

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